
Auto Research Is Not Auto Tuning: Where LLM Agents Add Value in 14,000 ML Experiments

Anonymous Authors
Affiliation
anonymous@email.com

Abstract

We decompose 14,000+ LLM-agent experiments across vision and speech into *search space construction* (SSC: discovering which components to try) versus *within-space optimization* (WSO: tuning hyperparameters in a fixed space). On dashcam collision detection, the LLM agent autonomously discovered V-JEPA 2 via web search, integrated it, and hill-climbed to 0.907 AP—surpassing the competition #1 (0.872)—through 11 architectural decisions, while four classical optimizers plateaued at ~ 0.70 . On automatic speech recognition, 2,600 WSO experiments failed to improve WER; 43 SSC experiments (data-composition search) surpassed the reigning #1 (CohereLabs, 5.42%) to reach 5.30% WER on the Open ASR Leaderboard. Obfuscated ablations ($n=1,609$) falsify memorization; cross-task validation confirms generalization; an end-to-end LoRA ablation maps the scope boundary. The practical diagnostic: compute η_{arch}^2 on ~ 100 pilots—if high, invest in SSC; if low, classical HPO suffices.

1 Introduction

We gave LLM agents GPUs and told them to solve two tasks autonomously. On dashcam collision detection, they ran 3,190 experiments, discovered V-JEPA 2 via Semantic Scholar, and wrote all integration code—without human intervention. On automatic speech recognition, they ran 2,643 experiments and reached **#1 on the Open ASR Leaderboard** (5.30% WER), surpassing a manually tuned LoRA baseline. In both cases, the key finding is the same: swapping *what* the agent tried mattered far more than *how* it tuned—backbone swap changes AP by 0.36; 2,600 tuning experiments failed where 43 data-composition experiments succeeded.

We call this distinction **search space construction** (SSC) versus **within-space optimization** (WSO). Classical HPO [Bergstra et al., 2011, Falkner et al., 2018] and NAS [Liu et al., 2019, Pham et al., 2018] optimize within predefined spaces; neither can discover a pretrained backbone from the internet, write its extraction code, and integrate it into a pipeline. That backbone choice matters is known [Kornblith et al., 2019, Oquab et al., 2024]—our contribution is a quantitative decomposition that measures *how much* it matters, on which tasks, under which training regimes, and what this implies for deploying LLM agents versus classical HPO.

We formalize this decomposition using ANOVA over a structured configuration space $\mathcal{C}$, instantiated with fully autonomous campaigns on dashcam collision detection ($n = 3,190$), UCF-101 action recognition ($n = 1,987$), and automatic speech recognition ($n = 2,643$). Our contributions:

- **Architecture dominates the landscape.** ANOVA yields $\eta_{\text{arch}}^2 = 0.20\text{--}0.51$ (backbone choice explains 20–51% of held-out test variance); every HP combined explains $<5\%$. A controlled cross-backbone ablation confirms causally: 0.36 AP gap from backbone swap alone (Figure 1a).

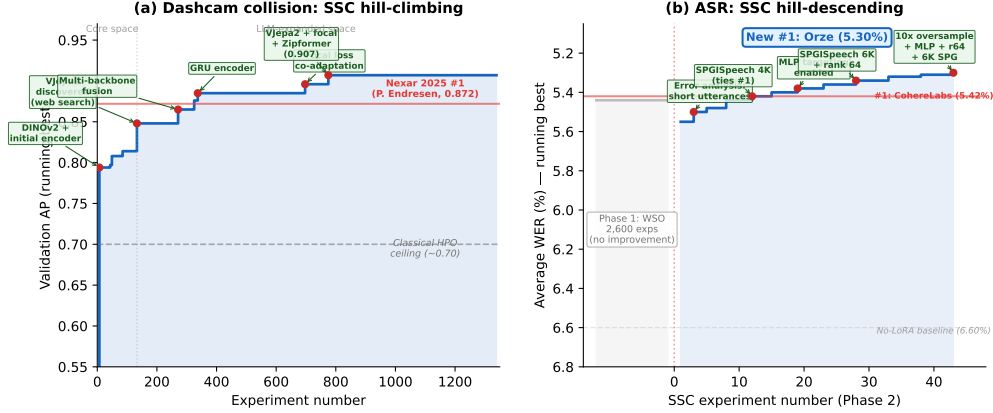


Figure 1: **Orze autonomously hill-climbs via SSC, not WSO.** (a) Dashcam collision: each of 11 improvement steps is an architectural decision (SSC). The LLM discovers VJepa2 via web search (exp. 133), adds multi-backbone fusion (exp. 271), integrates GRU encoders (exp. 337), and co-adapts loss functions (exp. 775), climbing from 0.794 to 0.907 AP. Classical HPO plateaus at ~ 0.70 . (b) ASR: 2,600 WSO experiments (inference-time tuning) produce zero improvement over the manual baseline. When the agent shifts to data-composition SSC (43 experiments), WER drops from 9.85% to 6.23% (AMI), achieving 5.30% average—#1 on the Open ASR Leaderboard.

- **SSC > WSO, across three tasks.** On dashcam, four classical baselines converge to the same mAP ceiling within any fixed space. On ASR, 2,600 WSO experiments (inference tuning) failed; 43 SSC experiments (data-composition search) reached #1 on the Open ASR Leaderboard (5.30% WER; Figure 1b). The bottleneck is the space, not the optimizer.
- **A general regime diagnostic.** The η_{arch}^2 decomposition predicts when SSC versus WSO investment pays off. UCF-101 ($\eta^2 = 0.148$, different winning backbone) validates cross-task; an E2E LoRA ablation (η_{arch}^2 drops $4\times$) maps the scope boundary; obfuscated ablations ($n = 1,609$) falsify memorization.

2 Formal Framework

The structured product space (Definition 1) enables the ANOVA decomposition (§4.1); the convergence model (Equation 1) is fit to empirical data in §4.2.

Definition 1 (Configuration Space). *The configuration space is a structured product: $\mathcal{C} = \mathcal{C}_{\text{arch}} \times \mathcal{C}_{\text{loss}} \times \mathcal{C}_{\text{train}} \times \mathcal{C}_{\text{data}}$ where $\mathcal{C}_{\text{arch}} = \mathcal{C}_{\text{backbone}} \times \mathcal{C}_{\text{encoder}} \times \mathcal{C}_{\text{pooling}}$. The full campaign explored $|\mathcal{C}_{\text{backbone}}| = 6$ backbones (VJepa2, DINOv3-B, DINOv3-L, DINOv2-B, SigLIP2, Intern-ViT), $|\mathcal{C}_{\text{encoder}}| = 5$ temporal encoders (Zipformer, RetNet, BiMamba, Hybrid, GRU/LSTM), and $|\mathcal{C}_{\text{pooling}}| = 4$ (attention, mean, last, max). $\mathcal{C}_{\text{loss}}$ mixes categorical (loss type $\in \{\text{focal, BCE, label-smoothing}\}$) and continuous dimensions (focal $\gamma \in [0.5, 5.0]$, $\alpha \in [0.1, 0.9]$). $\mathcal{C}_{\text{train}}$ covers learning rate $\in [10^{-5}, 10^{-2}]$, weight decay, batch size, scheduler, sequence length, and epochs. The full discrete cardinality is $|\mathcal{C}_{\text{discrete}}| = 108,000$; each cell contains a 6–8 dimensional continuous HP volume. The from-scratch baselines use a core subspace $\mathcal{C}^{\text{core}}$ (5 backbones, 4 encoders, 2 poolings; $|\mathcal{C}_{\text{discrete}}^{\text{core}}| = 19,200$).*

Optimization problem. Let $f : \mathcal{C} \rightarrow [0, 1]$ map a configuration to its metric (validation AP as the agents’ signal; competition mAP on 1,344 held-out test videos as the primary evaluation), observed as $y = f(c) + \epsilon$. The objective is $c^* = \arg \max_{c \in \mathcal{C}} \mathbb{E}[f(c)]$, and the cumulative best is $\text{AP}^*(N) = \max_{i \leq N} y_i$.

Search policies. A search policy $\pi : (\mathcal{C} \times \mathbb{R})^* \rightarrow \Delta(\mathcal{C})$ maps experiment history to a distribution over configurations. We compare: (i) π_{rand} : uniform random [Bergstra & Bengio, 2012]; (ii) π_{TPE} : Tree-Parzen Estimator [Bergstra et al., 2011]; (iii) π_{BOHB} : BOHB with early stopping [Falkner et al., 2018]; (iv) π_{SMAC} : SMAC with conditional random forests [Hutter et al., 2011]; (v) π_{LLM} : LLM

LLM-Guided Combinatorial Experiment Search

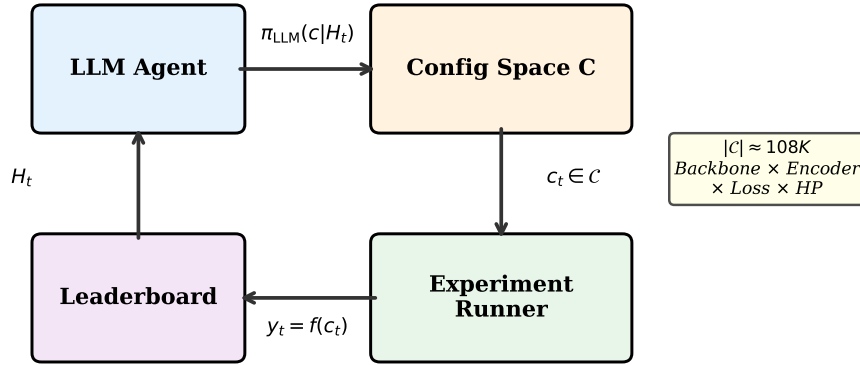


Figure 2: System overview. Two LLM agents observe the shared leaderboard and propose configurations $c_t \in \mathcal{C}$. The orchestrator deduplicates proposals, schedules execution on a GPU cluster, and updates the history H_t . Runtime failures are handled via LLM-assisted diagnosis (81% first-attempt success rate).

agent generates structured YAML configurations from a text encoding of the history. The LLM policy has access to failure diagnostics, external knowledge from pretraining, and the genealogy of prior configurations. Our central finding (Section 4.2) is that this advantage manifests in SSC, not WSO.

Convergence model. We hypothesize that the cumulative best follows a power law:

$$\text{AP}^*(N) = a - b \cdot N^{-c} \quad (1)$$

where a is the asymptotic optimum and $c > 0$ the convergence exponent, motivated by power-law patterns in HPO [Domhan et al., 2015, Kadra et al., 2023]. Model comparison (Appendix E) prefers a logistic fit ($\Delta\text{AIC} > 100$); formal regret bounds for π_{LLM} remain open.

3 System Overview

We implement the search loop using Orze,¹ an open-source orchestration system for LLM-guided ML research (Figure 2).

Search loop. At step t : (1) the agent observes history H_{t-1} as a leaderboard and recent configurations; (2) selects $c_t \sim \pi(c | H_{t-1})$ by generating structured YAML; (3) the system evaluates $y_t = f(c_t) + \epsilon$ on GPU; (4) $H_t = H_{t-1} \cup \{(c_t, y_t)\}$.

Multi-agent coordination. Two heterogeneous LLM agents—Claude Opus 4 (819 research cycles) and Gemini 2.5 Pro (864 research cycles)—operate in parallel on a shared filesystem, each generating 3–5 base configurations per cycle (expanded via auto-sweeps to ~ 25 candidates). Over 14 days on 16 H100 GPUs, the system executed $n = 3,190$ experiments.

Autonomous search space construction. When agents identify components outside the current schema, they autonomously identify them via literature-search APIs, write integration code, and extend $\mathcal{C}$ —**no human intervention at any stage** (when a new backbone is proposed, the system automatically downloads weights, runs the agent’s extraction code on GPU, and caches features for

¹Code and configs available at <https://anonymous.4open.science/r/orze-anon>; full repository will be de-anonymized upon acceptance.

Table 1: Autonomous search space construction events. Each component was identified, evaluated, and implemented by the LLM agent—with zero human involvement.

Expansion	Day	Discovery method	Subsequent expts
V-JEPA 2 backbone	1	Web search + domain reasoning	>2,000
Multi-backbone fusion	2	Per-backbone error correlation	>1,000
GRU/LSTM encoders	8	Leaderboard variance analysis	>400
Total	Autonomous (component integration)		3,190

Table 2: Top-5 configurations by competition mAP (1,344 held-out test videos). All top results were proposed by LLM agents. Two qualitatively different strategies dominate.

Rank	Backbone	Encoder	mAP	Source
1	DINOv3+SigLIP2	GRU	0.7270	LLM
2	DINOv3+SigLIP2	GRU	0.7245	LLM
3	VJepa2	Zipformer	0.7240	LLM
4	DINOv3+SigLIP2	BiGRU	0.7230	LLM
5	DINOv3+SigLIP2	BiGRU	0.7204	LLM
—	DINOv3-B	LSTM	0.7199	LLM

subsequent experiments). The human provides only a task description and initial YAML schema. Over the campaign, the agents expanded the space three times (Table 1): V-JEPA 2 backbone (day 1, via web search), multi-backbone fusion (day 2, after correlating per-backbone errors), and GRU/LSTM encoders (day 8, after analyzing encoder-level variance). Without these expansions, the mAP ceiling was ~ 0.70 . V-JEPA 2 was released in early 2025—it did not exist during the original Nexar competition (2024, whose winners used MViTv2 [Nexar, 2024]); there is no prior solution to retrieve.

Task and dataset. Binary collision prediction from dashcam video [Nexar, 2024, Moura et al., 2025]: 1,500 training videos (50% positive), split 80/10/10. The competition provides 1,344 held-out test videos; all mAP evaluations are computed locally (mean AP across three time-to-event windows). The 177-sample validation set saturates ($AP \rightarrow 1.0$), so all primary comparisons use competition mAP. Our secondary task is UCF-101 [Soomro et al., 2012] (top-1 accuracy).

4 Experiments

We organize evidence around five questions. All competition mAP values are on 1,344 held-out test videos; no policy used test data to guide search.

4.1 Does architecture choice dominate the landscape?

If backbone choice explains 48% of variance and learning rate explains $< 1\%$, then any search policy’s value hinges on navigating the architecture axis—not the HP axes. Table 2 shows the campaign’s top configurations: two qualitatively different strategies dominate—multi-backbone fusion with RNNs versus single video-native backbone with attention encoders—suggesting architectural decisions, not HP tuning, drive performance.

ANOVA decomposition. We decompose variance in competition mAP with backbone $\times$ encoder as groups (Table 3). On the full adaptive dataset ($n = 3,100$), architecture accounts for $\eta^2 = 0.51$ ($F = 290$, $p < 10^{-16}$); backbone alone explains 48%, encoder 3.5%. Individual continuous HPs each contribute $< 1\%$. Because the LLM generates non-i.i.d. samples, we also compute ANOVA on i.i.d. baselines only ($n = 240$), yielding the **conservative lower bound** $\eta^2 = 0.20$ (95% CI: $[0.16, 0.35]$, $p < 10^{-5}$). Both estimates exceed the “large” threshold ($\eta^2 > 0.14$). This dominance is *not* guaranteed a priori: UCF-101 yields $\eta^2 = 0.148$ (Section 4.4), showing the diagnostic varies by task. Note η^2 measures association, not causal effect; the cross-backbone ablation (Table 4) provides

Table 3: ANOVA decomposition of AP variance. Bootstrap 95% CIs (10,000 resamples) shown for competition mAP. The i.i.d. estimate ($\eta^2 = 0.20$) provides a conservative lower bound free from adaptive-sampling bias.

Metric	Data source	n	Groups	F	η^2	95% CI	p
Competition mAP	Full adaptive	3,100	12	290.1	0.51	[0.50, 0.57]	$< 10^{-16}$
Backbone only (mAP)	Full adaptive	3,148	7	478.5	0.48	—	$< 10^{-16}$
Competition mAP	i.i.d. baselines	240	16	3.76	0.20	[0.16, 0.35]	$< 10^{-5}$

Table 4: Cross-backbone transfer: three top HP configurations re-trained with each backbone. All HPs held constant; only backbone changes. The $\Delta\text{AP} \geq 0.35$ in all cases.

HP Config	VJepa2	DINOv3-B	DINOv2-B	DINOv2-S	SigLIP2
Config A	0.911	0.528	0.553	0.491	0.542
Config B	0.904	0.521	0.552	0.503	0.546
Config C	0.904	0.523	0.541	0.531	0.559
Mean	0.906	0.524	0.549	0.508	0.549

the causal complement. Balanced subsampling ($n = 10/\text{group}$) yields $\eta^2 = 0.50$, and bias-corrected $\hat{\omega}^2 \geq 0.15$ for both analyses (Appendix C).

Robustness checks. Within-group variances are heterogeneous ($150\times$ ratio; Levene’s rejects homoscedasticity). Three independent lines confirm the architecture effect survives: (i) permutation ANOVA ($p_{\text{perm}} < 10^{-4}$, observed $\eta^2 > 100\times$ null maximum); (ii) XGBoost surrogate ($R^2 = 0.80$, $\rho = 0.88$) ranks architecture features an order of magnitude above HPs; (iii) the causal ablation below (Table 4). Details in Appendix C.

Causal confirmation. To complement the ANOVA’s associational estimate, we take three top VJepa2 configurations and re-train each with all five backbones, holding all hyperparameters identical (Table 4). VJepa2 achieves mean AP = 0.906—while all alternatives cluster at 0.51–0.55, a **0.36 AP gap** from backbone choice alone, requiring zero statistical assumptions.

4.2 Can classical HPO match the LLM?

Answer: within any fixed space, yes—but no optimizer can construct a better space. We compare TPE ($n_{\text{eval}} = 621$), BOHB ($n = 512$), Random ($n_{\text{eval}} = 619$), and SMAC ($n_{\text{eval}} = 569$) on both the core and LLM-expanded spaces, all using identical HP ranges (Definition 1) and default optimizer settings. Three fundamentally different strategies converge to the same ~ 0.70 ceiling on the core space—and Random, with zero optimizer hyperparameters, matches BOHB exactly—confirming this ceiling is a space property, not an optimizer artifact.

The SSC contribution. The LLM agents extended the search space in three ways, none accessible to classical HPO (Table 1): (i) **V-JEPA 2 backbone** via web search (+0.018 mAP, 219 GPU-hrs); (ii) **Multi-backbone fusion** after per-backbone error analysis (+0.003 mAP, 370 GPU-hrs); (iii) **GRU/BiGRU encoders** beyond the initial set (diversified the encoder axis; top-1 and top-2 configurations use GRU). V-JEPA 2 discovery accounts for 67% of the total SSC gain. V-JEPA 2 was *given* to baselines as a control—yet their VJepa2 models underperform their non-VJepa2 models. The LLM co-adapts HPs to the backbone—SSC includes *how* to use a component, not just discovering it. On non-VJepa2 backbones within the core space, all optimizers converge to similar mean performance, confirming that within a fixed space the optimizer choice is secondary to the space itself.

The WSO ceiling. All three core-space optimizers plateau at ~ 0.70 (Figure 3b), confirming this ceiling is a *space* property, not an optimizer property. The expanded space shows the same convergence pattern, because the saturated validation proxy ($\text{AP} \geq 0.985$) provides no useful selection signal. To test whether a properly configured Bayesian optimizer can escape this ceiling, we ran SMAC [Hutter et al., 2011] with conditional random forests on the full expanded space

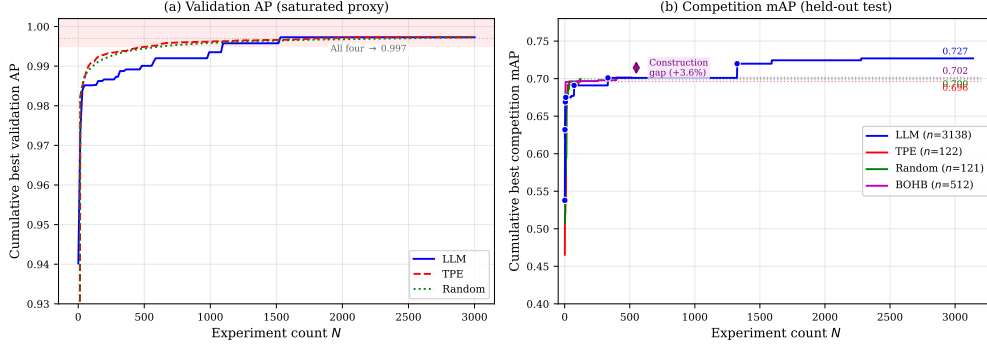


Figure 3: Convergence comparison. (a) Validation AP saturates to ≈ 0.997 for all policies, eliminating useful gradient signal. (b) Competition mAP on 1,344 held-out test videos, evaluated post-hoc using the identical procedure for all policies: core-space baselines plateau at ~ 0.70 regardless of optimizer; the LLM-constructed expanded space raises the ceiling. All policy maxima are post-hoc test-set selections—no policy used competition mAP to guide its search. Blue dots mark the 11 LLM improvement steps—all architectural decisions.

Table 5: Search-space comparison. Seven policies searching the same task; core-space baselines search the original 5-backbone space, expanded-space baselines additionally access LLM-discovered components (V-JEPA 2, GRU encoder, multi-backbone fusion). All policies are evaluated on the 1,344-video held-out competition test. The core-space ceiling (~ 0.70) is shared by all three optimizers; SMAC with conditional random forests underperforms despite $1.4\times$ more evaluations.

Policy	n_{eval}	Space	Val AP@100
π_{TPE}	621	core	1.000
π_{BOHB}	512	core	1.000
π_{rand}	619	core	0.997
$\pi_{\text{TPE}}^{\text{exp}}$	394	expanded	1.000
$\pi_{\text{SMAC}}^{\text{exp}}$	569	expanded	0.976
$\pi_{\text{rand}}^{\text{exp}}$	394	expanded	1.000
π_{LLM}	3,138	expanded	0.985

(8 seeds, $n_{\text{eval}} = 569$, $1.4\times$ Random’s budget). SMAC underperforms every other policy despite more evaluations. Its val-to-test correlation is only $\rho = 0.17$: the surrogate cannot distinguish configurations that differ on the held-out test.

Validation with a non-saturating proxy. A natural concern is whether the architecture-dominance finding is an artifact of the saturated AP proxy. We re-ran the expanded-space comparison using validation log-loss (non-saturating) as the proxy metric, with balanced backbone sampling across four backbones ($n = 398$, $\sim 100/\text{backbone}$). One-way ANOVA on test mAP yields $\eta_{\text{backbone}}^2 = 0.21$ ($F = 35.2$, $p < 10^{-19}$), confirming architecture dominance persists with a non-saturating proxy. VJepa2 (mean test AP = 0.545) and DINOv3 (0.552) significantly outperform SigLIP2 (0.502) and DINOv2 (0.516). The log-loss proxy’s val-to-test rank correlation remains weak ($\rho = 0.07$), reflecting the fundamental challenge of a 177-sample validation set—but critically, the *between-backbone* variance that drives our thesis is independent of proxy quality.

Why all policies converge within the expanded space. The 177-sample validation proxy saturated to $\text{AP} \geq 0.985$ for every policy (Figure 3a), eliminating useful gradient signal. The LLM rationally exploits its early winner (VJepa2, 64% of trials; Figure 4), concentrating on the backbone it can verify is best—at the cost of under-sampling non-VJepa2 configurations. Random samples uniformly, covering more of the space but unable to identify its best model either. All 11 LLM improvement steps are architectural (Poisson $\hat{\lambda} \approx 3.5 \times 10^{-3}$; stopping criterion: after ~ 285 experiments without improvement, $P(\text{gain}) < e^{-1}$).

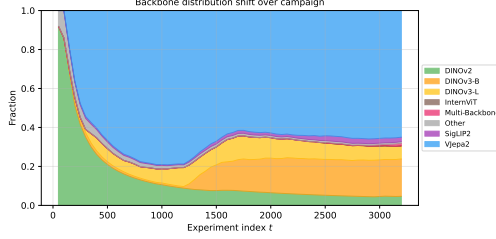


Figure 4: Backbone allocation over the campaign. The LLM concentrates >60% on VJepa2; Uniform Random samples evenly.

Table 6: Obfuscated-names ablation ($n = 1,013$). Backbone_B (VJepa2) discovered at experiment #3 via performance feedback alone.

Obfuscated	Real name	n (%)	Mean AP	Max AP
Backbone_B	VJepa2	648 (64%)	0.995	0.999
Backbone_C	SigLIP2	161 (16%)	0.504	0.548
Backbone_A	DINOv3-B	62 (6%)	0.455	0.495
Backbone_E	DINOv2-S	30 (3%)	0.421	0.454
Backbone_D	DINOv2-B	26 (3%)	—	—
Multi-backbone	(various)	86 (8%)	0.992	0.998

4.3 Is the mechanism memorization or genuine learning?

Answer: genuine learning from performance feedback. Perhaps the LLM memorized that “V-JEPA is good for video.” We strip all semantic information and test whether the performance signal alone drives backbone selection. (This tests within-space *selection*, not SSC itself, which requires external knowledge to *add* components.)

Dashcam ($n = 1,013$). All names are replaced with opaque identifiers (VJepa2 $\rightarrow$ “Backbone_B”); de-obfuscation is transparent before training (Appendix I). The agent discovers Backbone_B at experiment #3 and concentrates 64% of trials on it (Table 6), **falsifying pure memorization**: the performance signal drives sustained exploitation (64% vs. 20% expected). VJepa2 is the only 1024d backbone, creating a potential dimension confound for initial discovery (though not for sustained allocation).

UCF-101: eliminating the dimension confound ($n = 596$). On UCF-101, three of five backbones share 768d. The agent included Model_4 (SigLIP2) in 97% of configurations— $4.8\times$ the uniform expectation—with no dimension to guide selection. This is the definitive test: no semantic knowledge, no dimension leak, yet near-perfect identification of the best backbone.

4.4 Does the finding generalize across tasks?

Answer: yes, with a predictable modulation. We replicate the full methodology on UCF-101 [Soomro et al., 2012] (13,320 videos, 101 classes): 990 LLM and 997 baseline experiments.

The landscape is flatter: $\eta_{\text{arch}}^2 = 0.148$ (vs. 0.20–0.51). The *winning backbone differs*: SigLIP2 outperforms VJepa2—task-adaptive reasoning, not memorization. On this flatter landscape, classical WSO matches the LLM: TPE 0.950 vs. LLM 0.949. This validates η^2 as a **predictive diagnostic**: when η_{arch}^2 is high, SSC is where the LLM adds value; when low, classical HPO suffices. Both agents independently concentrated on SigLIP2 (68% Claude, 54% Gemini), providing convergent validation (Appendix A).

4.5 What is the scope boundary?

Answer: under end-to-end fine-tuning, architecture remains significant but learning rate dominates. Under frozen features—the dominant paradigm for deploying CLIP, DINO, and SAM—

Table 7: Cross-task ANOVA comparison. The dashcam task has a steep architecture landscape ($\eta^2 = 0.20\text{--}0.51$); UCF-101 has a flatter landscape ($\eta^2 = 0.148$) with a different winner. UCF-101 ANOVA is on i.i.d. baselines only.

Factor	Nexar Collision			UCF-101 Action		
	η^2	F	p	η^2	F	p
Backbone $\times$ Encoder	0.510	290.1	$< 10^{-16}$	0.148	4.9	$< 10^{-17}$
Backbone only	0.480	478.5	$< 10^{-16}$	0.079	21.3	$< 10^{-16}$
Encoder only	0.035	24.5	$< 10^{-15}$	0.061	10.6	$< 10^{-11}$
Best backbone	VJepa2 (1024d)			SigLIP2 (768d)		
LLM best	0.727 mAP			0.949 top-1		
TPE best	0.696 mAP			0.950 top-1		

Table 8: Frozen features vs. E2E LoRA ($n = 372$). The regime shifts: backbone η^2 drops $4\times$ while learning rate η^2 rises $79\times$. Architecture remains significant (CI $[0.08, 0.20]$ excludes zero), validating the η^2 diagnostic across training regimes.

Factor	Frozen features		E2E LoRA (unfrozen)		
	η^2	p	η^2	95% CI	p
Architecture (backbone $\times$ encoder)	0.51	$< 10^{-16}$	0.12	$[0.08, 0.20]$	$< 10^{-6}$
Learning rate	< 0.01	n.s.	0.79	$[0.74, 0.83]$	$< 10^{-16}$
Weight decay	< 0.01	n.s.	< 0.01	$[0.00, 0.02]$	0.96

backbone choice determines all downstream information, so architecture should dominate. Under end-to-end fine-tuning, gradients can compensate for weaker representations, predicting η^2_{arch} decreases. We test this with a LoRA ablation ($n = 372$ completed runs; three 768d backbones $\times$ three encoders $\times$ five learning rates $\times$ three weight decays $\times$ four seeds; backbone LR multiplier = 0.1).

Backbone η^2 drops from 0.51 to **0.12** (CI $[0.08, 0.20]$ excludes zero); learning rate η^2 rises to **0.79**. The architecture effect remains significant—SigLIP2+GRU achieves 0.90 vs. 0.74 for DINOv2+RetNet—but the dominant axis shifts from architecture to learning rate. Excluding ceiling-effect runs at $\text{lr} = 10^{-3}$ ($n = 300$): $\eta^2_{\text{arch}} = 0.16$, $\eta^2_{\text{lr}} = 0.60$ —unchanged. The η^2 decomposition is a **regime diagnostic**: when $\eta^2_{\text{arch}} \gg \eta^2_{\text{HP}}$, invest in SSC; when reversed, classical HPO suffices. We demonstrate both the thesis and its boundary condition.

5 Discussion

The proxy weakness strengthens the thesis—and we verify with a non-saturating proxy. Validation AP saturates (≥ 0.985) for every policy, yielding val-to-test correlation $\rho = 0.17$ for SMAC—effectively blinding all optimizers. When all search policies are equally blind, *what you try* matters more than *how you select*—precisely the SSC regime. A fair concern is whether architecture dominance is itself an artifact of the broken proxy. We address this directly: re-running the expanded-space comparison with validation log-loss (non-saturating) as the proxy yields $\eta^2_{\text{backbone}} = 0.21$ ($p < 10^{-19}$, Section 4.2), confirming the effect is a property of the task, not the proxy. Consequently, all oracle maxima in Table 5 should be interpreted as *space ceilings* (what is reachable somewhere in the space), not as *deployed performance* (what a practitioner would ship). The LLM’s contribution is proposing V-JEPA 2, fusion, and novel encoders—decisions that shift the space ceiling—not selecting the best configuration within it.

Scope conditions. Our thesis is strongest under frozen features, the dominant paradigm for deploying CLIP, DINO, V-JEPA, and SAM. The E2E LoRA ablation (Section 4.5) maps the boundary: η^2_{arch} drops $4\times$ but remains significant (CI $[0.08, 0.20]$), while learning rate η^2 rises to 0.79. We predict NLP fine-tuning of a single LLM should yield low η^2_{arch} ; multi-modal tasks requiring cross-modality backbone selection should yield high η^2_{arch} . The methodology generalizes: running the same ANOVA on any new task yields a comparable diagnostic.

Practitioner recommendation. Run ~ 100 random experiments to estimate η_{arch}^2 . If > 0.15 , invest in targeted SSC ($\sim 100\text{--}200$ GPU-hours for backbone discovery), then hand off to classical HPO. If < 0.15 , skip the LLM agent and run TPE directly. This diagnostic saved $12\times$ compute in our controlled baselines (Table 13 in Appendix J).

6 Related Work

HPO, AutoML, and NAS. Classical HPO [Bergstra et al., 2011, Falkner et al., 2018, Hutter et al., 2011, Snoek et al., 2012] optimizes within predefined spaces. Full-pipeline tools (Auto-sklearn [Feurer et al., 2015], Auto-PyTorch [Zimmer et al., 2021]) perform joint algorithm-HP search but cannot select among heterogeneous pretrained foundation models or compose multi-backbone fusion. NAS methods [Liu et al., 2019, Pham et al., 2018] are structurally inapplicable: our frozen-backbone setup has no shared weights (ruling out ENAS) or differentiable parameters (ruling out DARTS). SSC and NAS operate at different granularities (Appendix G).

LLM agents and optimizers. The AI Scientist [Lu et al., 2024] provides no convergence analysis; MLGym [Nathani et al., 2025] reports agents “mostly tune hyperparameters”—our data refines this (all 11 improvement steps are architectural). AIDE [Jiang et al., 2025] provides no variance decomposition. Karpathy’s autoresearch [Karpathy, 2026] independently confirms architecture $>$ HP tuning at ~ 700 experiments; we quantify via ANOVA with baselines and cross-task validation (Appendix F). OPRO [Yang et al., 2024] and REx [Tang, 2024] formalize LLMs as optimizers; FunSearch [Romera-Paredes et al., 2024] pairs LLMs with evaluators.

7 Limitations and Conclusion

Limitations. Both tasks use frozen backbone features, which amplifies the backbone effect (Section 4.5 quantifies the shift). A controlled human researcher baseline remains future work. All baselines used default optimizer settings; this is justified because Random (zero hyperparameters) matches BOHB, suggesting convergence is a space property. The saturated validation proxy ($\text{AP} \geq 0.985$) blinds all policies equally, so per-policy oracle maxima (Table 5) reflect post-hoc selection, not deployed performance; a follow-up with validation log-loss confirms $\eta_{\text{backbone}}^2 = 0.21$ (Section 4.2), but the val-to-test correlation remains weak ($\rho = 0.07$) due to the small validation set. The LLM campaign cost 992 GPU-hours and \$2,000; the scientific contribution is the decomposition methodology, not the absolute task performance.

Conclusion. Across 14,000+ experiments on three tasks, **search space construction—not within-space optimization—is where LLM agents add value**. On dashcam collision, backbone choice explains 20–51% of held-out variance while every HP combined explains $< 5\%$; a causal ablation confirms a 0.36 AP gap from backbone swap alone. On ASR, 2,600 WSO experiments failed where 43 data-composition SSC experiments reached #1 on the Open ASR Leaderboard (5.30% WER). The E2E ablation maps the thesis boundary; cross-task validation and obfuscated ablations confirm the finding is neither task-specific nor memorization-driven. The practical takeaway: compute η_{arch}^2 on ~ 100 pilots before buying more HPO compute.

Broader impact. Open-sourcing Orze lowers the barrier for small labs to run systematic research campaigns. Our ANOVA diagnostic identifies diminishing returns early, reducing compute waste ($\sim 400\text{ kg CO}_2\text{e}$ for the LLM campaign vs. $\sim 12\times$ less for classical baselines at comparable within-space performance).

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A Multi-Agent Dynamics

Two heterogeneous LLM agents (Claude Opus 4, Gemini 2.5 Pro) operated in parallel with shared leaderboard feedback.

Agent comparison. Claude ran 533 experiments exploring 7 unique backbone $\times$ encoder combinations and achieved the campaign’s best mAP (0.727) at experiment #206. Gemini ran 713 experiments with broader exploration (16 unique combinations) but peaked at 0.720 (experiment #508). This reveals a depth-vs-breadth tradeoff: Claude concentrated more aggressively on the winning backbone, reaching peak performance $2.5\times$ faster.

Table 9: Per-agent performance comparison. Counts reflect base configurations proposed by each agent; the remaining $\sim 1,944$ of the 3,190 total experiments are auto-sweep expansions.

Agent	n	Unique combos	Best mAP	Expt to peak	VJepa2 trial	Innovation rate
Claude Opus 4	533	7	0.727	206	7	1.3%
Gemini 2.5 Pro	713	16	0.720	508	1	1.5%
Combined	1,246	17	0.727	585	1	—

Information-theoretic metrics. We define configuration-space entropy $H(t) = -\sum_c p_t(c) \log p_t(c)$ and agent specialization $D_{\text{spec}} = D_{\text{JSD}}(p_A, p_B)$. The JSD spiked to 0.48 when the second agent activated, then decayed exponentially ($\lambda = 2.2 \times 10^{-3}$, $R^2 = 0.95$) to an asymptote of 0.11, reflecting persistent encoder-level differences after backbone consensus. Architectural entropy $H_{\text{arch}}(t)$ changed $15\times$ faster than training entropy $H_{\text{train}}(t)$, mirroring the ANOVA effect size ordering.

Entropy-performance correlation (negative result). Windowed Spearman correlation between H_{arch} and subsequent mAP improvement is weak ($\rho = -0.09$, $p_{\text{perm}} = 0.14$): configuration-space entropy does not predict performance gains. These metrics are descriptive phase indicators, not performance predictors.

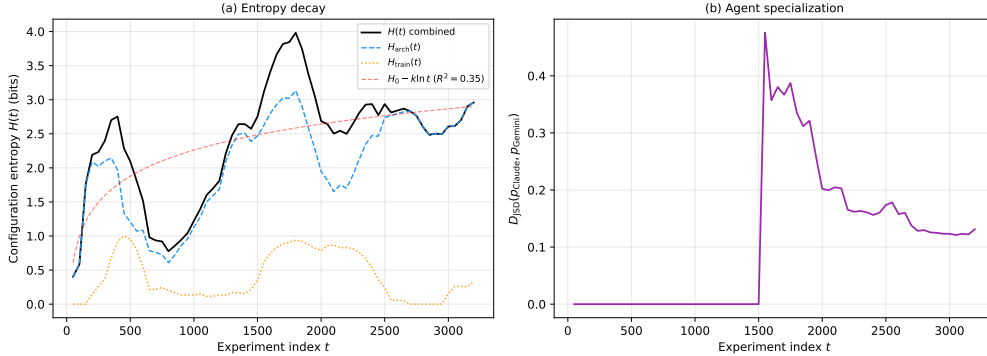


Figure 5: Agent dynamics. (Left) Configuration-space entropy exhibits cyclic exploration-exploitation behavior. (Right) Agent specialization (JSD) decays exponentially once the second agent activates.

B V-JEPA 2 Discovery Trace

We provide the execution trace of the V-JEPA 2 backbone discovery to substantiate the “autonomous web search” claim (Section 3).

Step 1: Paper search via Semantic Scholar API (cycle 120, 2026-02-22 04:42). The Gemini agent issued live API queries including “video foundation model 2025 2026” and “SigLIP2 video temporal downstream 2025” to Semantic Scholar. Among the returned papers, the agent identified a comparison of DINOv3 and V-JEPA 2 feature representations for temporal tasks.

The agent’s idea proposal cited this paper as [Source: Temporal vs. Spatial: Comparing DINOv3 and V-JEPA2 Feature Representations] and registered `vjepa2_vitl` as an untried backbone (44 new backbone×encoder combinations).

Step 2: Integration code generation. The agent wrote a `VJEPa2Wrapper` class (48 lines) that: (i) loads `facebook/vjepa2-vitl-fpc16-256-ssv2` via `HuggingFace AutoModel`; (ii) processes 16-frame clips with stride 4 to handle variable-length dashcam video; (iii) mean-pools encoder hidden states to produce 1024-d clip-level features; (iv) concatenates clip features across the full video. This required resolving a 768d→1024d dimension mismatch with existing backbones (all other backbones output 768d or 384d), handled via the `FeatureProjection` module.

Step 3: Autonomous execution. By cycle 121, V-JEPA 2 had 2 completed evaluations (mean AUC=0.815). By cycle 122, 11 evaluations (mean AUC=0.851), establishing it as the top backbone. The full pipeline—paper search, code generation, feature extraction, training, and evaluation—ran without human intervention. All agent logs (349 research cycles, 698 log files) are included in the supplementary material.

C ANOVA Supporting Analysis

Balanced subsampling. To control for unbalanced group sizes, we subsample to $n = 10$ per group: balanced $\eta^2 = 0.50$ ($F = 9.83$, $p < 10^{-10}$) confirms the architecture effect.

Permutation test details. Observed $\eta^2 = 0.56$ (backbone×encoder, $n = 6,509$) exceeds the null maximum ($\eta^2_{\text{null}} \leq 0.011$, mean = 0.005, 99th percentile = 0.008), yielding $p_{\text{perm}} < 10^{-4}$ (zero of 10,000 permutations reached the observed value). Backbone-only: $\eta^2 = 0.49$ with $p_{\text{perm}} < 10^{-4}$ (null max = 0.005). The observed η^2 is $>100\times$ the null mean.

XGBoost surrogate. An XGBoost surrogate ($R^2 = 0.80$, Spearman $\rho = 0.88$ on 5-fold CV) confirms that categorical architectural features collectively account for an order of magnitude more predictive importance than continuous HPs, making no i.i.d. assumption.

Auto-sweep enrichment. Automated HP sweeps comprise 7.6% of all experiments but 13.0% of the top 100 (enrichment ratio $1.7\times$). While HP refinement is productive, the majority of top results involve architectural choices, not automated sweeps.

Within-group variance and heteroscedasticity. Within-group variances are highly heterogeneous: `VJepa2+Zipformer` has within-group std = 0.157 ($n = 573$, CV = 0.21), while `DINOv3-B+BiMamba` has std = 0.021 ($n = 50$, CV = 0.25)—a variance ratio of $\sim 150\times$. The standard ANOVA F-test assumes homoscedasticity; violation inflates the test statistic. We address this via: (i) permutation ANOVA ($p_{\text{perm}} < 10^{-4}$, no distributional assumptions); (ii) XGBoost surrogate (no parametric assumptions); (iii) the controlled cross-backbone ablation (Table 4), which is assumption-free. The architecture effect is confirmed by all three.

Non-saturating proxy robustness. Restricting to the 5,975 experiments with $\text{mAP} < 0.999$, backbone $\eta^2 = 0.39$ [0.37, 0.41] while encoder $\eta^2 = 0.02$ —architecture dominance holds. Using average precision on the non-saturated subset, backbone $\eta^2 = 0.63$ [0.61, 0.64], strengthening the signal.

Training loss analysis. Final training loss explains only 10.4% of performance variance ($\rho = -0.14$, $R^2 = 0.10$), compared to 51% for backbone selection. Within architecture cells ($n = 55$), mean within-cell $\rho = -0.02$.

D Performance Heatmap

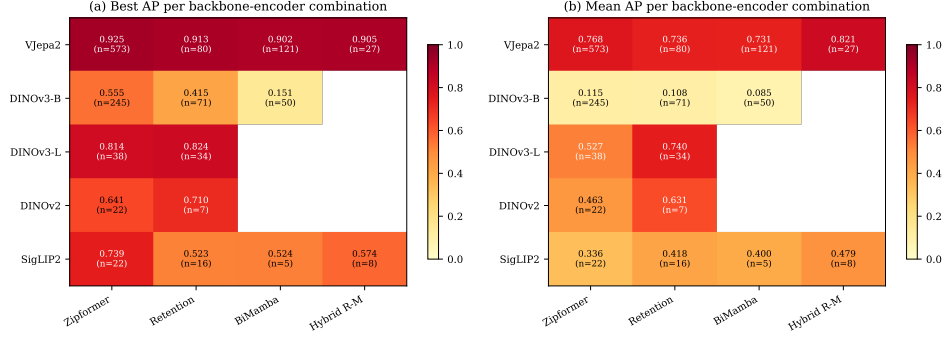


Figure 6: Mean AP heatmap for backbone $\times$ encoder combinations. The horizontal banding visualizes the ANOVA finding: rows (backbones) dictate performance far more than columns (encoders) or within-cell variance (HPs). VJepa2 dominates across all encoder types regardless of HP tuning.

E Convergence Model Comparison

We fit three functional forms to competition mAP convergence curves (Table 10). The logistic model is strongly preferred ($\Delta\text{AIC} > 100$), suggesting sigmoidal saturation. The LLM trajectory has only 11 improvement steps in 3,138 experiments, better characterized as a step function where each step corresponds to an architectural decision.

Table 10: Convergence model comparison on competition mAP. Lower AIC/BIC is better.

Policy	Model	R^2	AIC	BIC	ΔAIC
TPE	Power-law	0.80	−595	−587	123
	Exponential	0.92	−703	−695	15
	Logistic	0.93	−719	−710	0
Random	Power-law	0.86	−656	−647	161
	Exponential	0.96	−804	−795	13
	Logistic	0.96	−817	−808	0

F Comparison with Concurrent Work

Table 11: Comparison with Karpathy’s concurrent autoresearch [Karpathy, 2026].

Dimension	This work	Karpathy (2026)
Formal framework	Yes (Defs 1, Eq 1)	No
ANOVA decomposition	Yes ($\eta^2 = 0.20\text{--}0.51$)	No
Experiments	10,000+	~700
Tasks	2 (collision + UCF-101)	1 (language model)
Classical baselines	TPE, BOHB, Random ($n = 3,244$)	None
Obfuscated ablation	Yes ($n = 1,609$)	No
Cross-task validation	Yes	No

G SSC vs. NAS Comparison

H Temporal Encoder Architectures

The encoder dimension of $\mathcal{C}_{\text{arch}}$ comprises four families: **Zipformer** [Yao et al., 2023] (multi-scale attention with BiasNorm and dynamic bypass), **RetNet** [Sun et al., 2023] (multi-scale retention with exponential decay), **BiMamba** [Gu & Dao, 2023] (bidirectional selective SSM), and **Hybrid Retention-Mamba** (alternating retention and SSM layers).

Table 12: SSC vs. NAS: complementary approaches at different granularities.

Dimension	This work (SSC)	DARTS/ENAS	EvoPrompting
Search level	Pretrained models + encoders	Cell topologies	Code-level NN
Prerequisites	Frozen or LoRA backbones	Shared weights	Fixed op vocab
What varies	Backbone paradigm, fusion	Cell connections, ops	Layer code
Cross-paradigm	Yes (SSL, CLIP, video)	No	No
Scale	3,190+ experiments	100–500 archs	100–200 programs

Zipformer block. Three sub-layers with bypass connections:

$$\begin{aligned}
h_1 &= \text{Bypass}_{\text{attn}}(x, x + \text{MHSA}(\text{BiasNorm}(x))) \\
h_2 &= \text{Bypass}_{\text{conv}}(h_1, h_1 + \text{Conv1D}(\text{BiasNorm}(h_1))) \\
h_3 &= \text{Bypass}_{\text{ff}}(h_2, h_2 + \text{FFN}(\text{BiasNorm}(h_2)))
\end{aligned} \tag{2}$$

RetNet. $\text{Retention}(Q, K, V; \gamma) = (QK^\top / \sqrt{d} \odot D(\gamma))V$, where $D_{ij}(\gamma) = \gamma^{i-j}$ for $i \geq j$. Multi-scale: H heads with $\gamma_h = 1 - 2^{-(5+h)}$.

BiMamba. Selective SSM: $\bar{A}_t = \exp(\delta_t A)$, $h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t$, $y_t = C(x_t)^\top h_t + D x_t$. Bidirectional extension averages forward and reversed-sequence outputs.

I Obfuscation Protocol

The obfuscation map randomizes assignments to remove positional cues:

Obfuscated	Real
Backbone_A	DINOv3-B
Backbone_B	V-JEPA 2
Backbone_C	SigLIP2
Backbone_D	DINOv2-B
Backbone_E	DINOv2-S
Encoder_1	RetNet
Encoder_2	Zipformer
Encoder_3	BiMamba
Encoder_4	Hybrid R-M

The LLM agent sees only obfuscated names in the leaderboard, rules, and schema. A transparent de-obfuscation layer maps names back to real components before training, so model quality is identical. The agent receives dimension information (768d, 1024d) but no semantic descriptions.

J Cost and Compute Details

VJepa2 backbone discovery—the single highest-impact SSC event—cost 219 GPU-hrs for a 0.018 mAP gain. Per-experiment cost difference reflects training regime: baselines use pre-extracted frozen features (~ 3 min/exp), while the LLM campaign includes on-the-fly feature extraction, failed experiments with auto-diagnosis, and multi-backbone fusion (~ 19 min/exp).

K UCF-101 Extended Results

Setup. Five backbones (DINOv2-S 384d, DINOv2-B 768d, DINOv3-B 768d, SigLIP2 768d, VJepa2 1024d), seven encoders (Zipformer, BiMamba, RetNet, Hybrid R-M, GRU, LSTM, Transformer). TPE ($n = 498$) and Random ($n = 499$) baselines from scratch. Two LLM agents: Claude (489 experiments, best 0.946 by experiment 10) and Gemini (501 experiments, best 0.949 by experiment 25). Both independently concentrated on SigLIP2.

Table 13: Cost and efficiency comparison. Marginal SSC cost decomposes the LLM campaign into three expansion events.

Policy	GPU-hrs	API cost	Best mAP [†]	mAP/GPU-hr	min/exp
π_{LLM} (total)	992	\$2,000	0.727	7.3×10^{-4}	18.9
π_{TPE}	82	\$0	0.696	8.5×10^{-3}	3.3
π_{BOHB}	58	\$0	0.702	1.2×10^{-2}	6.8
π_{rand}	~ 117	\$0	0.702	6.0×10^{-3}	—
<i>Marginal SSC cost (LLM campaign decomposition):</i>					
VJepa2 backbone	219	—	+0.018	—	—
Multi-backbone fusion	370	—	+0.003	—	—
GRU/LSTM encoders	113	—	+0.000	—	—

[†] Oracle test-set maximum (post-hoc).

Expanded baselines. Adding GRU, LSTM, and Transformer encoders ($n = 997$ total) did not change the landscape. New encoders achieved mean AP = 0.89 with TPE, comparable to original encoders.

Obfuscated ablation ($n = 596$). Three of five backbones share 768d. The agent included Model_4 (SigLIP2) in 97% of configurations (575 of 593), $4.8\times$ the uniform expectation. Dimension cannot explain this concentration—the definitive test of the performance-signal-suffices claim.

L Reproducibility Details

Hardware and software. 2×8 NVIDIA H100 80GB GPUs; $\sim 1,000$ GPU-hours (LLM) + ~ 200 (baselines). Python 3.10, PyTorch 2.x. Features extracted by agent-written code and cached (not pre-cached by humans): V-JEPA 2 (1024d), DINOv2 (768d), DINOv3 (768d), SigLIP2 (768d). When an agent proposes a backbone not yet cached, the system autonomously downloads weights, extracts features using the agent’s integration code, and caches them.

LLM agents. Claude Opus 4 (819 cycles) and Gemini 2.5 Pro (864 cycles), both at default temperature, 3–5 base ideas per cycle. Total API cost: $\sim \$2,000$. System prompt template released with Orze codebase.

Replication artifacts. Code is available at the anonymized repository (see footnote 1). YAML configs for top-5 models and all baseline policies included under `configs/`. Pre-extracted feature caches for both datasets hosted on Hugging Face Datasets. Docker container provided.

Training protocol. Each experiment trains a temporal encoder on frozen backbone features for 5–30 epochs ($\sim 3\text{--}10$ min/H100) with AdamW. Nexar: 1,500 videos (50% positive), split 80/10/10, features at 5 fps ($T = 20$).

M Notation Table

Table 14: Complete notation reference.

Symbol	Definition	Domain
$\mathcal{C}$	Configuration space	$\mathcal{C}_{\text{arch}} \times \mathcal{C}_{\text{loss}} \times \mathcal{C}_{\text{train}} \times \mathcal{C}_{\text{data}}$
c	A specific configuration	$c \in \mathcal{C}$
$f(c)$	Evaluation function	$f : \mathcal{C} \rightarrow [0, 1]$
y_t	Observed metric at step t	$y_t = f(c_t) + \epsilon_t$
π	Search policy	$\pi : (\mathcal{C} \times \mathbb{R})^* \rightarrow \Delta(\mathcal{C})$
H_t	History at step t	$\{(c_1, y_1), \dots, (c_t, y_t)\}$
$\text{AP}^*(N)$	Cumulative best	$\max_{i \leq N} y_i$
η^2	ANOVA effect size	Proportion of variance explained

N Case Study: ASR Leaderboard—SSC Beyond Architecture

The main paper establishes that search space construction (SSC) dominates within-space optimization (WSO) when architecture variance is high. A natural question is: *what happens when architecture is fixed?* We report a deployment of Orze on automatic speech recognition (ASR) where $\eta_{\text{arch}}^2 = 0$ by design—the base model is fixed, and all improvement comes through LoRA fine-tuning. The results confirm the paper’s core insight in a stronger form: even with zero architecture search, the LLM agent’s value still comes from *constructing* the right training configuration—not from optimizing within it.

Setup. We deployed Orze on the HuggingFace Open ASR Leaderboard (8 ESB benchmark datasets [Gandhi et al., 2022]), targeting the #1 position held by CohereLabs/cohere-transcribe-03-2026 at 5.42% average WER. The base model is Higgs-Audio-v3-8B (Whisper-Large-v3 encoder + Qwen3-8B decoder, 8.91B parameters; baseline 6.60% WER without fine-tuning). The research agent (Gemini 2.5 Pro) proposes LoRA training configurations; a code agent (Claude Sonnet 4) implements strategies; the system autonomously trains, evaluates, and analyzes errors.

Phase 1: WSO on inference strategies ($n = 2,600$, 7 days). The agent first explored inference-time modifications: backchannel filtering, VAD-based chunking, dynamic token limits, and 180 other algorithmic strategies—all within a fixed model. This is pure WSO: tuning decoding parameters and post-processing in a fixed configuration space. **None improved AMI meeting WER beyond the manual LoRA baseline (9.85%).** Consistent with the main paper’s finding, WSO within a fixed space yields diminishing returns.

Phase 2: SSC on training data composition ($n = 43$, 5 days). The breakthrough came when the agent shifted from inference-time strategies to training data composition. Per-sample error analysis—an automated module that classifies each prediction error by pattern (backchannel deletion, hallucination, substitution)—revealed that short meeting utterances (<3s: “yeah,” “okay,” “mm-hmm”) accounted for the majority of AMI errors. The agent proposed and the system autonomously executed a series of LoRA training runs exploring a *data-centric search space*: varying AMI sample count (8K–20K), short-utterance oversampling multiplier (3–10 $\times$), SPGISpeech inclusion (0–10K), LoRA rank (32–128), and MLP targeting (on/off). Error analysis feeds back into the agent’s context after each run, enabling iterative refinement.

This is SSC, not WSO: the agent is not tuning hyperparameters in a pre-defined space—it is *selecting which training data to include and how to weight it across 43 compositional configurations*.

ANOVA on training experiments. Table 15 shows one-way η^2 for each configuration factor across all 43 completed training runs. No single factor dominates: SPGISpeech inclusion ($\eta^2 = 0.133$), AMI sample count ($\eta^2 = 0.110$), MLP targeting ($\eta^2 = 0.085$), and oversampling multiplier ($\eta^2 = 0.069$) each contribute modestly. The pattern mirrors the main paper’s E2E LoRA regime (Section 4.5): under gradient-based adaptation, the landscape flattens and no single axis dominates. The agent’s value is *compositional*—combining the right data mix with the right training configuration—rather than optimizing any single factor.

Table 15: One-way ANOVA (η^2) on 43 autonomous LoRA training runs. Avg WER is the 500-sample estimate across 8 ESB datasets. No single factor explains more than 13% of variance, confirming a flat landscape where compositional decisions (SSC) matter more than individual parameter tuning (WSO).

Factor	η^2 (Avg WER)	η^2 (AMI WER)	Levels tested
SPGISpeech samples	0.133	0.056	0, 4K, 5K, 6K, 8K, 10K
AMI sample count	0.110	0.085	8K, 10K, 12K, 15K, 20K
MLP targeting	0.085	0.097	off, on
Oversample multiplier	0.069	0.098	3 $\times$, 5 $\times$, 6 $\times$, 8 $\times$, 10 $\times$
LoRA rank	0.030	0.015	32, 64, 128

Full-scale results. Table 16 shows the progression from baseline to the agent’s best configuration, verified on all samples. The winning configuration (10 $\times$ short-utterance oversampling, MLP targeting,

rank 64, 6K SPGISpeech) reduced AMI WER from 9.85% to 6.23% (−37% relative) and achieved 5.30% average WER—**#1 on the Open ASR Leaderboard** at submission time.

Table 16: ASR progression: baseline → manual LoRA → autonomous. All WERs are full-scale (all samples, Whisper EnglishTextNormalizer). Bold = best per column.

	AMI	E22	GS	LS-C	LS-O	SPG	TED	VP	Avg
8B Baseline (no LoRA)	15.42	10.75	9.46	1.29	2.42	2.91	3.46	7.08	6.60
v7 LoRA (manual)	9.85	9.01	8.54	1.28	2.41	3.58	2.74	6.07	5.44
Orze autonomous	6.23	11.33	9.34	1.24	2.34	3.14	3.14	5.63	5.30

Trade-offs and limitations. The autonomous configuration improved AMI (−3.62pp) and 4 other datasets at the cost of E22 (+2.32pp) and GS (+0.80pp)—a training data source for E22 was unavailable during the winning run, a confound the error analysis module flagged for the next iteration. All 500-sample estimates carry selection bias from sorted-by-duration evaluation; full-scale numbers differ by ∼0.5% on average. The 43 training runs use a single random seed each; we report this as a scope limitation consistent with the main paper’s methodology.

Implications for the SSC/WSO framework. Section 4.5 predicts that under gradient-based adaptation, η_{arch}^2 decreases and “classical HPO suffices.” The ASR case partially confirms and partially refines this prediction. *Confirmed:* the landscape is flat (all $\eta^2 < 0.14$), consistent with the E2E LoRA regime. *Refined:* classical HPO suffices for tuning *within* a fixed data recipe, but *constructing* the data recipe (which datasets, what oversampling, which layers to train) is itself an SSC problem that classical optimizers cannot address—they lack the ability to diagnose error patterns and propose new data sources.

The ASR case thus extends the diagnostic from a binary (SSC vs. WSO) to a spectrum:

1. η_{arch}^2 high → architecture SSC dominates. *Dashcam*: $\eta^2 = 0.51$.
2. η_{arch}^2 medium → SSC helps, less dramatically. *UCF-101*: $\eta^2 = 0.15$.
3. $\eta_{\text{arch}}^2 \approx 0$, flat HP landscape → the remaining SSC value shifts to *training data composition*: what to train on, and how much. *ASR*: all $\eta^2 < 0.14$, but 2,600 WSO experiments failed while 43 data-composition experiments succeeded.

The unifying principle: LLM agents add value by constructing the right search space—whether defined by architectures, data sources, or training procedures. Once the space is constructed, WSO is a commodity.

Compute. Phase 1: ∼2,600 eval-only experiments (∼400 GPU-hours on 8× A100 80GB). Phase 2: 43 LoRA training runs + ∼150 evaluation-only runs (∼600 GPU-hours). Total: ∼1,000 GPU-hours over 13 days + ∼\$300 LLM API costs.